Experiment 4

Title: JavaScript Program using Variables, Conditions, and Loops

Problem Statement:
Create a JavaScript program that takes user input and displays output using conditions and loops.

Create a web application that:

1. Takes the **number of subjects** from the user.
2. Uses a **loop** to take marks for each subject.
3. Calculates:
    - Total Marks
    - Average Marks
4. Uses **if-else conditions** to display:
    - Grade
    - Pass/Fail Result
5. Displays the result dynamically on the webpage.

Objective:

To develop a web-based application using **HTML, CSS, and JavaScript** that:

- Takes user input
- Uses **loops**
- Uses **conditional statements**
- Displays formatted output on the webpage

Covers Syllabus:

- Variables & data types
- Operators
- Loops & conditionals

Expected Outcome:

Correct processing and display of user input.

Tools: HTML, JavaScript, CSS

Experiment 5

Title: Array and Object Operations using JavaScript

Problem Statement:
Store student details using arrays and objects.
Display, filter, and count records using loops and array methods.

Create a web-based system where:

1. User enters:
    - Employee Name
    - Employee ID
    - Salary
    - Department
2. Each employee record should be stored as an **object** inside an **array**.
3. Provide buttons to:
    - Display all employees
    - Filter employees with salary greater than ₹50,000
    - Calculate total salary payout
    - Calculate average salary
    - Count employees in a specific department
4. Display all results dynamically on the webpage

Objective:

Develop a web application using **HTML, CSS, and JavaScript** that:

- Stores employee details using **objects inside an array**
- Displays all employees
- Filters employees based on salary
- Calculates total and average salary
- Counts employees department-wise

Covers Syllabus:

- Arrays & objects
- for…in, for…of
- map / filter (basic level)

Expected Outcome:
Student data processed and displayed correctly.

Tools: JavaScript, html, css

Experiment 6

Title: DOM Manipulation and Event Handling using JavaScript

Problem Statement:
Create buttons and input fields that dynamically change webpage content using JavaScript events.

Create a web page that includes:

1. A text input field to enter a message.
2. Buttons to:
    - Change heading text
    - Change background color
    - Change font size
    - Show/Hide paragraph
    - Reset page
3. All actions must be handled using **JavaScript event listeners** such as:
    - onclick
    - onmouseover
    - onchange
    - addEventListener()
4. The webpage content must change dynamically without reloading.

Objective:

To develop a web application using **HTML, CSS, and JavaScript** that:

- Uses different **JavaScript events**
- Dynamically changes webpage content
- Modifies styles using DOM manipulation
- Uses event handling functions

Covers Syllabus:

- DOM manipulation
- Event handling
- Functions

Expected Outcome:
Dynamic updates without page reload.

Tools: HTML, JavaScript, CSS